

Pratham Arora

[✉ pratham3992@gmail.com](mailto:pratham3992@gmail.com) [in LinkedIn](#) [G GitHub](#) [Portfolio](#)

EDUCATION

Plaksha University

B.Tech. in Computer Science & AI — CGPA: 8.19 / 10.0

Mohali, India
Expected Aug. 2026

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, OpenCV, SAM, GANs
Languages: Python (NumPy, Pandas), C++, MATLAB, SQL, JavaScript
Cloud & MLOps: Azure, Docker, FastAPI, RAG Pipelines, Git, Linux, Firebase
Tools: Jupyter, VS Code, Google Colab, Unity (VR Integration)

INDUSTRY EXPERIENCE

AI Intern

June 2025 – July 2025

Cotality | Kolkata

- Engineered a Generative AI system for automated code documentation, utilizing **Abstract Syntax Tree (AST)** analysis across 15+ languages.
- Deployed a **Vector Database** (Azure Cosmos DB) for RAG implementation, optimizing retrieval latency and reducing token usage costs by 90%.

SDE Intern (Robotics)

June 2024 – July 2024

Orangewood Labs | Noida

- Extended the **RoboGPT** framework, integrating LLM-based planning with robotic control loops for complex manipulation tasks.
- Refined the Computer Vision stack for 6D object pose detection, significantly reducing error margins in robotic arm trajectory planning.

SELECTED PROJECTS

Pre-eclampsia Discourse Analysis | NLP, VADER, LDA, Data Engineering

Aug. 2025 – Dec. 2025

- Processed 22,000+ Reddit interactions using **VADER** sentiment analysis and **Latent Dirichlet Allocation (LDA)** for topic modeling.
- Identified a "Buffer Effect" (19.4% sentiment shift) and quantified a 35.8x post-pandemic volume increase in health communities.

VR LLM Conversational Agent | Unity, Gemini 2.5 Flash, HCI, RAG

Aug. 2025 – Dec. 2025

- Developed an LLM-powered embodied conversational agent for VR using **Gemini 2.5 Flash** and cloud speech services.
- Designed and conducted a within-subjects user study ($N = 18$) to evaluate verbal, visual, and gestural latency feedback mechanisms.
- Implemented multi-modal interaction strategies to analyze their impact on perceived animacy, immersion, and social presence.
- Performed statistical analysis (Friedman, Wilcoxon, paired t -tests) on Godspeed and custom survey metrics.
- Authored a full research paper on latency perception and embodied feedback in immersive AI agents.

Kelp Forest Segmentation | Python, PyTorch, U-Net, Satellite Imagery

Jan. 2024 – May 2024

- Designed a **U-Net** architecture for semantic segmentation of kelp forests in noisy satellite imagery, achieving a performance rank in the **Top 100 out of 600+** competitors.
- Engineered a pre-processing pipeline using custom CNN filters to remove atmospheric noise and improve label quality.

AI-Powered Resume & Interview Coach | Next.js, Gemini API, Prompt Engineering

Jan. 2025 – April 2025

- Engineered an interactive "Interview GPT" module using **Gemini API**, designing custom system prompts to simulate rigorous technical interviews and provide real-time DSA feedback.
- Developed a generative resume engine that utilizes LLMs to autonomously draft professional content based on raw user inputs.

LEADERSHIP

Head of Tech

April 2025 – Present

Athleda Sports Society | Plaksha Univ.

- Overseeing technical architecture and deployment for the university's sports platform.

Head of Web Development

Feb. 2024 – Feb. 2025

LEAP.AI@Plaksha | Plaksha Univ.

- Led the web team for the AI club's official website and mentored juniors in full-stack best practices.